Kinematics Formula Sheet

Physics • Class 11 / JEE Main & Advanced / NEET

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1. Motion in a Straight Line (1D)

1st Equation:	$v = u + at$
2nd Equation:	$s = ut + \frac{1}{2}at^2$
3rd Equation:	$v^2 = u^2 + 2as$
Dist. in n^{th} sec:	$s_n = u + \frac{a}{2}(2n - 1)$
Average Velocity:	$v_{\text{avg}} = \frac{u+v}{2} = \frac{\Delta x}{\Delta t}$
Displacement:	$s = \left(\frac{u+v}{2}\right) t$

2. Free Fall & Vertical Motion (g)

Max Height (u upwards):	$H = \frac{u^2}{2g}$
Time of Ascent:	$t_a = \frac{u}{g}$
Total Flight Time:	$T = \frac{2u}{g}$
Velocity at Ground (h):	$v = \sqrt{2gh}$
Drop from Rest (t):	$h = \frac{1}{2}gt^2$
Galileo's Odd Ratio:	$1 : 3 : 5 : 7 : \dots$

3. Oblique Projectile Motion (Angle θ with Horizon)

Time of Flight:	$T = \frac{2u \sin\theta}{g} = \frac{2u_y}{g}$
Maximum Height:	$H_{\max} = \frac{u^2 \sin^2\theta}{2g} = \frac{u_y^2}{2g}$
Horizontal Range:	$R = \frac{u^2 \sin 2\theta}{g} = \frac{2u_x u_y}{g}$
Max Range ($\theta = 45^\circ$):	$R_{\max} = \frac{u^2}{g} = 4H_{\max}$
Trajectory Equation:	$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$
Range Form:	$y = x\tan\theta\left(1 - \frac{x}{R}\right)$
Velocity at time t:	$\vec{v} = (u\cos\theta)\hat{i} + (u\sin\theta - gt)\hat{j}$
Speed at height h:	$v = \sqrt{u^2 - 2gh}$

4. Relative Velocity & Vectors

Relative Velocity:	$\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B$
Relative Accel.:	$\vec{a}_{A/B} = \vec{a}_A - \vec{a}_B$
River (Shortest Time):	$t_{\min} = \frac{d}{v_{br}}$ (at 90°)
River (Zero Drift):	$\sin\theta = \frac{v_r}{v_{br}}$ ($v_{br} > v_r$)
Rain-Man Problem:	$\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$

5. Circular Motion Kinematics

Angular Velocity:	$\omega = \frac{d\theta}{dt} = \frac{v}{r}$
Angular Accel.:	$\alpha = \frac{d\omega}{dt} = \frac{a_t}{r}$
Centripetal Accel.:	$a_c = \frac{v^2}{r} = \omega^2 r$
Tangential Accel.:	$a_t = \frac{dv}{dt} = \alpha r$
Net Acceleration:	$a_{\text{net}} = \sqrt{a_c^2 + a_t^2}$
Rotational Eq.:	$\omega = \omega_0 + \alpha t$

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